■ B.3.7 Iterators

`{`*imax*`}`	iterate *imax* times
`{`*i*`,` *imax*`}`	*i* goes from *1* to *imax* in steps of *1*
`{`*i*`,` *imin*`,` *imax*`}`	*i* goes from *imin* to *imax* in steps of *1*
`{`*i*`,` *imin*`,` *imax*`,` *di*`}`	*i* goes from *imin* to *imax* in steps of *di*
`{`*i*`,` *imin*`,` *imax*`}, {`*j*`,` *jmin*`,` *jmax*`}`	*i* goes from *imin* to *imax*, and for each value of *i*, *j* goes from *jmin* to *jmax*

Iterator notation.

Iterator notation is used in such functions as `Sum`, `Table`, `Do` and `Range`.

The iteration parameters *imin*, *imax* and *di* do not need to be integers. The variable *i* is given a sequence of values starting at *imin*, and increasing in steps of *di*, stopping when the next value of *i* would be greater than *imax*. The iteration parameters can be arbitrary symbolic expressions, so long as `(`*imax*`-`*imin*`)/`*di* is a number.

When several iteration variables are used, the limits for the later ones can depend on the values of earlier ones.

The variable *i* can be any symbolic expression; it need not be a single symbol. However, the expression given for *i* must have no value assigned to it at the time when the iteration function is evaluated.

The procedure for evaluating iteration functions is described on page 676.